



Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

LEARNER GUIDE

For

Microsoft Certified Azure AI Engineer Associate AI-102 Training

TGS Ref No: TGS-2023036651

Conducted by
Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

Version 2.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	13 Jul 2026	Initial 10-lab AI-102 courseware release	Tertiary Infotech Academy Pte Ltd
2.0	26 Aug 2026	Current Microsoft Foundry terminology; AI-102 retirement transition; technical evidence model; 10 detailed self-contained labs.	Tertiary Infotech Academy Pte Ltd

TABLE OF CONTENTS

01	How to Use This Guide
02	Before You Start
03	Lab 01 - Plan, Manage, Monitor, and Secure an Azure AI Solution
04	Lab 02 - Responsible AI, Content Safety, Governance
05	Lab 03 - Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG
06	Lab 04 - Agentic AI, Foundry Agent Service, Multi-Agent Concepts
07	Lab 05 - Computer Vision, Custom Vision, Video Insights
08	Lab 06 - NLP, Language, Speech, Translation
09	Lab 07 - Custom Language Models and Question Answering
10	Lab 08 - Azure AI Search, Knowledge Mining, Vector Search
11	Lab 09 - Document Intelligence and Content Understanding
12	Lab 10 - AI-102 Capstone and Course Review
13	Quick Command Reference
14	Assessment Flow and Support

How to Use This Guide

Use the trainer deck to understand mechanisms and decisions. Use this guide for the detailed click paths, commands, code, expected results, diagnostics, and evidence required in each lab.

Authority	Current reference
Approved course	Microsoft Certified Azure AI Engineer Associate AI-102 Training · TGS-2023036651 · 2 days / 16 hours
Course page	https://www.tertiarycourses.com.sg/wsqa-microsoft-certified-azure-ai-engineer-associate-ai-102-training.html
Final AI-102 blueprint	Skills measured 23 Dec 2025; exam retired 30 Jun 2026
Current certification path	AI-103 skills measured from 16 Apr 2026
LMS	https://lms-tms.tertiaryinfotech.com/

Before You Start

- Use the Azure subscription or lab environment supplied by the trainer.
- Install current Azure CLI/Python tooling only when the lab requires it.
- Authenticate with Microsoft Entra ID where supported; never save live keys in the repository.
- Use synthetic data and remove training resources after evidence has been captured.
- For every lab, retain request IDs, deployment/model versions, screenshots or JSON outputs, and a short interpretation.

Lab 01 - Plan, Manage, Monitor, and Secure an Azure AI Solution

Source folder: labs/lab-01-plan-manage-secure-azure-ai-solution

Objectives

- Select appropriate Azure AI services.
- Plan resource deployment and endpoints.
- Identify authentication, keys, monitoring, and cost controls.
- Prepare an AI solution architecture.

Scenario

A company wants a customer support AI platform with document search, answer generation, sentiment analysis, speech transcription, and image processing. You must plan the Azure AI resources.

Steps

1. Map Requirements to Services

2. Plan Foundry Resources

Document:

Hub or project name
Region
Model deployment
Default endpoint
Connected data source
Responsible AI settings

3. Plan Security

Record:

- Role-based access control.
- Managed identities.
- Key protection.
- Private endpoint or network controls.
- Logging and diagnostics.
- Secrets storage.

4. Plan Monitoring and Cost

Document:

Token or transaction usage
Latency
Errors
Content safety events
Budget alerts
Diagnostic logs

5. Draw Architecture

Use diagrams.net:

Application -> API layer -> Foundry/Azure AI services -> Search index/storage -> Monitoring

Validation

You should have a service map, security plan, monitoring plan, and architecture diagram.

Checkpoint Questions

1. Why is service selection part of AI engineering?
2. Why should keys be protected?
3. What should be monitored in an AI service?
4. What is the role of responsible AI planning?

Course Focus

AI engineering starts with selecting the right service, securing it, monitoring it, and planning for cost and responsible use.

Acceptance Evidence

Evidence pack: resource topology, RBAC assignment, private connectivity decision, successful SDK call, and an alert driven by a real metric.

Troubleshooting Checklist

5. Capture the full error and request/operation ID before retrying.
6. Confirm the endpoint, API/model version, region, identity, and RBAC scope.
7. Validate payload/schema fields and input content type.
8. Check quota, rate limits, network resolution, private connectivity, and service status.
9. Record the corrective action and rerun the acceptance check.

Lab 02 - Responsible AI, Content Safety, Governance

Source folder: labs/lab-02-responsible-ai-content-safety-governance

Objectives

- Apply responsible AI principles.

- Configure content safety concepts.
- Explain content filters, blocklists, prompt shields, and harm detection.
- Design a responsible AI governance framework.

Scenario

The support AI assistant may receive abusive user input, produce inaccurate answers, or expose sensitive data. You must design responsible AI controls.

Steps

1. Review Responsible AI Principles

Write one control for:

Fairness
Reliability and safety
Privacy and security
Inclusiveness
Transparency
Accountability

2. Define Content Safety Controls

Document:

- Harm categories.
- Severity levels.
- Content filters.
- Blocklists.
- Prompt shields.
- Groundedness checks.
- Human escalation rules.

3. Create a Safety Policy

Write:

Allowed use:
Disallowed use:
Blocked content:
Human review trigger:
Audit log requirement:
Escalation owner:
Review frequency:

4. Test Governance Scenarios

Decide what should happen when:

10. User asks for harmful instructions.
11. Model produces low-confidence answer.
12. Prompt contains personal data.
13. Retrieved document is outdated.
14. User asks for legal advice.

5. Create an Incident Plan

Define how to report, investigate, remediate, and prevent repeat AI safety incidents.

Validation

You should have responsible AI controls, safety policy, scenario decisions, and incident plan.

Checkpoint Questions

15. What are content filters?
16. What is a prompt shield?
17. Why is human escalation important?
18. Who is accountable for AI output?

Course Focus

Responsible AI is implemented through policy, configuration, monitoring, human review, and governance.

Acceptance Evidence

Evidence pack: risk register, test dataset, per-category confusion matrix, filter configuration, exception workflow, and incident rehearsal result.

Troubleshooting Checklist

19. Capture the full error and request/operation ID before retrying.
20. Confirm the endpoint, API/model version, region, identity, and RBAC scope.
21. Validate payload/schema fields and input content type.
22. Check quota, rate limits, network resolution, private connectivity, and service status.
23. Record the corrective action and rerun the acceptance check.

Lab 03 - Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG

Source folder: labs/lab-03-generative-ai-foundry-openai-prompt-flow-rag

Objectives

- Plan a generative AI solution with Microsoft Foundry.
- Select and deploy a model conceptually.
- Design prompt templates and prompt flow.
- Implement a RAG pattern conceptually.
- Evaluate model and flow outputs.

Scenario

The company wants a grounded assistant that answers questions from internal support articles and generates draft replies.

Steps

1. Select a Model

Record:

Model name
Use case fit
Context window
Latency
Cost
Safety settings
Deployment option

2. Design Prompt Templates

Create templates for:

- 24. Direct answer.
- 25. Answer with sources.
- 26. Escalation when context is missing.

3. Plan Prompt Flow

Draw:

```
User question -> classify intent -> retrieve context -> prompt template -> model -> safety check -> response
```

4. Design RAG

Document:

- Source documents.
- Cleaning.
- Chunking.
- Embeddings.
- Vector store.
- Retrieval filters.
- Citation format.
- Evaluation method.

5. Evaluate Outputs

Score:

Validation

You should have model selection notes, prompt templates, prompt flow diagram, RAG design, and evaluation table.

Checkpoint Questions

- 27. What is RAG?
- 28. Why are prompt templates useful?
- 29. What does tracing help debug?
- 30. How can feedback improve a generative AI solution?

Course Focus

Generative AI solutions need grounded data, evaluation, safety settings, monitoring, and feedback loops.

Acceptance Evidence

Evidence pack: chunk sample, index schema, hybrid query trace, cited answer, groundedness score, latency, and a negative test with no supporting evidence.

Troubleshooting Checklist

- 31. Capture the full error and request/operation ID before retrying.
- 32. Confirm the endpoint, API/model version, region, identity, and RBAC scope.
- 33. Validate payload/schema fields and input content type.
- 34. Check quota, rate limits, network resolution, private connectivity, and service status.
- 35. Record the corrective action and rerun the acceptance check.

Lab 04 - Agentic AI, Foundry Agent Service, Multi-Agent Concepts

Source folder: labs/lab-04-agentic-ai-foundry-agent-service

Objectives

- Explain agent roles and use cases.
- Plan Foundry Agent Service resources.
- Design custom agent tools and workflows.
- Review multi-agent orchestration concepts.

Scenario

The support AI assistant needs to look up order status, search support documents, draft a reply, and escalate complex cases to a human agent.

Steps

1. Define Agent Use Case

Write:

```
Agent goal:
User types:
Allowed tools:
Disallowed actions:
Human escalation trigger:
Success metric:
```

2. Plan Agent Resources

Document:

- Foundry project.
- Model deployment.
- Search tool.
- Function or API tools.
- Authentication.
- Logging and trace settings.

3. Design Tool Calls

Create a table:

4. Design Orchestration

Draw:

```
User request -> agent plan -> tool call -> observation -> model reasoning -> response -> feedback
```

5. Multi-Agent Review

Explain when separate agents might be useful for retrieval, billing, technical support, and escalation review.

Validation

You should have agent use case, resource plan, tool table, orchestration diagram, and multi-agent notes.

Checkpoint Questions

- 36. What is an AI agent?
- 37. Why do agents need tool permissions?
- 38. What is orchestration?
- 39. Why should autonomous actions be constrained?

Course Focus

Agentic solutions combine models, tools, workflows, monitoring, safety controls, and human escalation.

Acceptance Evidence

Evidence pack: agent definition/version, tool schemas, least-privilege identity, approval gate, trace showing tool call/result, budget exhaustion test, and handoff test.

Troubleshooting Checklist

- 40. Capture the full error and request/operation ID before retrying.
- 41. Confirm the endpoint, API/model version, region, identity, and RBAC scope.
- 42. Validate payload/schema fields and input content type.
- 43. Check quota, rate limits, network resolution, private connectivity, and service status.
- 44. Record the corrective action and rerun the acceptance check.

Lab 05 - Computer Vision, Custom Vision, Video Insights

Source folder: labs/lab-05-computer-vision-custom-vision-video

Objectives

- Analyze images with Azure AI Vision concepts.
- Select visual features for image processing requirements.
- Compare image classification and object detection.
- Plan custom vision training.
- Explain video insight scenarios.

Scenario

The company wants to inspect product photos, detect damaged items, read text from images, and extract insights from product demo videos.

Steps

1. Map Vision Requirements

2. Plan an Image Request

Document:

Image source
Visual features
Language
Expected response fields
Error handling
Privacy considerations

3. Custom Vision Plan

Choose:

Document image labels, training images, evaluation metrics, publishing, and consumption.

4. Video Insights Plan

Explain how Video Indexer can extract:

- Transcript.
- Keywords.
- Faces or speakers where appropriate.
- Topics.
- Scenes.

5. Responsible AI Review

Document consent, retention, face analysis sensitivity, and human review needs.

Validation

You should have vision mapping, request plan, custom vision plan, video insight notes, and responsible AI review.

Checkpoint Questions

45. What is OCR?
46. How is object detection different from classification?
47. What does a custom vision model require?
48. Why is video analysis sensitive?

Course Focus

Vision solutions require selecting visual features, model type, evaluation approach, and responsible data handling.

Acceptance Evidence

Evidence pack: sample request/response, confidence policy, confusion matrix, model version, difficult-image tests, and privacy/retention decision.

Troubleshooting Checklist

49. Capture the full error and request/operation ID before retrying.
50. Confirm the endpoint, API/model version, region, identity, and RBAC scope.
51. Validate payload/schema fields and input content type.
52. Check quota, rate limits, network resolution, private connectivity, and service status.
53. Record the corrective action and rerun the acceptance check.

Lab 06 - NLP, Language, Speech, Translation

Source folder: labs/lab-06-nlp-language-speech-translation

Objectives

- Analyze and translate text.
- Extract key phrases, entities, PII, language, and sentiment.
- Process speech with speech-to-text and text-to-speech.
- Plan translation and SSML use.

Scenario

The support platform must analyze customer messages, detect PII, translate text, transcribe calls, and generate spoken responses.

Steps

1. Map Language Tasks

2. Design Text Analysis Pipeline

Input text -> language detection -> PII detection -> sentiment -> entities -> key phrases
-> route case

3. Design Speech Pipeline

Audio input -> speech to text -> intent or keyword recognition -> response -> text to speech

4. SSML Review

Explain how SSML can control voice, pitch, rate, pauses, and pronunciation.

5. Translation Review

Document:

- Text translation.
- Document translation.
- Speech-to-text translation.
- Speech-to-speech translation.
- Human review for critical messages.

Validation

You should have language task mapping, text pipeline, speech pipeline, SSML notes, and translation notes.

Checkpoint Questions

54. What is entity recognition?
55. Why detect PII before storing text?
56. What is SSML?
57. When should translation outputs be reviewed?

Course Focus

Language and speech solutions combine analysis, privacy controls, translation, and application integration.

Acceptance Evidence

Evidence pack: multilingual samples, PII span/redaction, recognition reasons, SSML output, word-error-rate sample, and human review for domain terms.

Troubleshooting Checklist

58. Capture the full error and request/operation ID before retrying.
59. Confirm the endpoint, API/model version, region, identity, and RBAC scope.
60. Validate payload/schema fields and input content type.
61. Check quota, rate limits, network resolution, private connectivity, and service status.
62. Record the corrective action and rerun the acceptance check.

Lab 07 - Custom Language Models and Question Answering

Source folder: labs/lab-07-custom-language-question-answering

Objectives

- Design custom language understanding.
- Define intents, entities, and utterances.
- Plan custom question answering.
- Create a multilingual question answering strategy.

Scenario

The support assistant must recognize customer intents and answer frequently asked questions from approved sources.

Steps

1. Define Intents and Entities

Create a table:

2. Plan Training Data

Document:

- Number of example utterances.
- Balanced examples across intents.
- Entity labeling rules.
- Test set.
- Ambiguous utterances.

3. Plan Custom Question Answering

Record:

Knowledge sources
Q&A pairs
Alternate phrasing
Chit-chat
Multi-turn prompts
Publishing target
Fallback answer

4. Multilingual Strategy

Explain whether to translate sources, create separate projects, or translate user input before answering.

5. Backup and Recovery

Document export, versioning, backup, and recovery of language projects.

Validation

You should have intent/entity table, training plan, question answering plan, multilingual strategy, and backup notes.

Checkpoint Questions

63. What is an intent?
64. What is an entity?
65. Why add alternate phrasing?

66. Why should language projects be backed up?

Course Focus

Custom language solutions need well-designed examples, evaluation, publishing, and lifecycle management.

Acceptance Evidence

Evidence pack: label guide, train/test split, confusion matrix, deployment name, fallback threshold, Q&A source citation, export, and recovery rehearsal.

Troubleshooting Checklist

- 67. Capture the full error and request/operation ID before retrying.
- 68. Confirm the endpoint, API/model version, region, identity, and RBAC scope.
- 69. Validate payload/schema fields and input content type.
- 70. Check quota, rate limits, network resolution, private connectivity, and service status.
- 71. Record the corrective action and rerun the acceptance check.

Lab 08 - Azure AI Search, Knowledge Mining, Vector Search

Source folder: labs/lab-08-ai-search-knowledge-mining-vector-search

Objectives

- Design an Azure AI Search solution.
- Define indexes, data sources, indexers, and skillsets.
- Explain semantic and vector search.
- Plan knowledge store projections.

Scenario

The company wants searchable support documents, enriched with extracted text, entities, key phrases, and vector embeddings for RAG.

Steps

1. Design an Index

Define fields:

```
id
title
content
category
source_url
last_updated
security_label
embedding
```

2. Plan Data Sources and Indexers

Document:

- Source storage.
- Supported document formats.
- Indexer schedule.
- Change detection.
- Error handling.

3. Define a Skillset

Include:

- OCR.
- Language detection.
- Entity extraction.
- Key phrase extraction.
- Custom skill.
- Embedding generation.

4. Query Design

Write examples for:

Keyword search
Filter by category
Sort by last updated
Wildcard query
Semantic ranking
Vector search
Hybrid search

5. Knowledge Store Projections

Explain file, object, and table projections and when enriched output should be stored.

Validation

You should have index design, data source/indexer plan, skillset plan, query examples, and knowledge store notes.

Checkpoint Questions

72. What is an indexer?
73. What is a skillset?
74. What is vector search?
75. Why use semantic ranking?

Course Focus

Knowledge mining combines source data, enrichment, indexing, querying, and optional vector retrieval for AI applications.

Acceptance Evidence

Evidence pack: index schema, data source/indexer/skillset JSON, successful run history, judged queries, hybrid trace, security-filter negative test, and knowledge-store projection.

Troubleshooting Checklist

76. Capture the full error and request/operation ID before retrying.
77. Confirm the endpoint, API/model version, region, identity, and RBAC scope.
78. Validate payload/schema fields and input content type.
79. Check quota, rate limits, network resolution, private connectivity, and service status.
80. Record the corrective action and rerun the acceptance check.

Lab 09 - Document Intelligence and Content Understanding

Source folder: labs/lab-09-document-intelligence-content-understanding

Objectives

- Use prebuilt document extraction concepts.
- Plan custom Document Intelligence models.
- Explain composed models.
- Design Content Understanding extraction workflows.

Scenario

The company processes invoices, contracts, support screenshots, audio notes, and product videos. It wants to extract structured information from multiple content types.

Steps

1. Map Document Scenarios

2. Custom Document Model Plan

Document:

Document types
Sample count
Fields to extract
Labels
Training split
Evaluation metric
Publish plan

3. Content Understanding Plan

Explain extraction from:

- Documents.
- Images.
- Video.
- Audio.

Include summarization, classification, attributes, entities, tables, and images.

4. Human Review Rules

Define confidence thresholds and when extracted data requires review.

5. Integration Flow

Draw:

Upload content -> extract text/entities/tables -> validate -> store structured data -> search/report

Validation

You should have scenario mapping, custom model plan, content understanding plan, review rules, and integration flow.

Checkpoint Questions

81. When should you use a prebuilt model?
82. What is a composed model?
83. Why are confidence scores useful?
84. What content types can Content Understanding process?

Course Focus

Information extraction solutions need model selection, training data, confidence handling, and integration design.

Acceptance Evidence

Evidence pack: analyzer/model ID, sample result JSON, source polygons/spans, field-level metrics, review threshold, failure sample, and downstream search/workflow record.

Troubleshooting Checklist

- 85. Capture the full error and request/operation ID before retrying.
- 86. Confirm the endpoint, API/model version, region, identity, and RBAC scope.
- 87. Validate payload/schema fields and input content type.
- 88. Check quota, rate limits, network resolution, private connectivity, and service status.
- 89. Record the corrective action and rerun the acceptance check.

Lab 10 - AI-102 Capstone and Course Review

Source folder: labs/lab-10-ai102-capstone-course-review

Objectives

- Design an end-to-end Azure AI solution.
- Map requirements to Microsoft Foundry and Azure AI services.
- Include security, monitoring, responsible AI, and cost controls.
- Build a personal review plan.

Scenario

You must design a production-ready customer support AI platform that supports grounded chat, document search, sentiment analysis, speech transcription, image OCR, document extraction, and human escalation.

Steps

1. Build a Service Map

2. Draw the Architecture

Include:

User application
API layer
Foundry project
Model deployment
AI Search index
Storage
Language/Speech/Vision/Document services
Monitoring
Human review queue

3. Add Governance Controls

Document:

- RBAC.
- Managed identity.
- Key protection.

- Prompt filters.
- Content safety.
- Logging and monitoring.
- Budget alerts.
- Data retention.
- Human escalation.

4. Rate Skill Confidence

5. Clean Up Resources

If you created resources:

```
az group delete --name rg-ai102-lab --yes --no-wait
```

Confirm with your instructor before deleting shared resources.

6. Create a 7-Day Review Plan

- 90. Day 1: Planning, security, monitoring, responsible AI.
- 91. Day 2: Generative AI, prompt flow, RAG.
- 92. Day 3: Agents and orchestration.
- 93. Day 4: Vision and video.
- 94. Day 5: NLP, speech, translation, question answering.
- 95. Day 6: AI Search, Document Intelligence, Content Understanding.
- 96. Day 7: Capstone review and mistakes log.

Validation

You should have a service map, architecture diagram, governance checklist, confidence matrix, cleanup plan, and review plan.

Checkpoint Questions

- 97. Which service should power grounded document search?
- 98. When should a human review AI output?
- 99. What is the difference between RAG and fine-tuning?
- 100. Which AI workload do you need to review most?

Course Focus

Azure AI engineering is about integrating the right AI services into secure, monitored, responsible, production-ready solutions.

Acceptance Evidence

Capstone evidence: architecture and data flow, ADR, threat/risk model, evaluation report, SLO dashboard, cost estimate, deployment/rollback plan, runbook, and improvement backlog.

Troubleshooting Checklist

- 101. Capture the full error and request/operation ID before retrying.
- 102. Confirm the endpoint, API/model version, region, identity, and RBAC scope.
- 103. Validate payload/schema fields and input content type.
- 104. Check quota, rate limits, network resolution, private connectivity, and service status.
- 105. Record the corrective action and rerun the acceptance check.

Quick Command Reference

Command / item	Purpose
az login --use-device-code	Authenticate Azure CLI without embedding credentials
az account show --output table	Confirm active tenant/subscription
python -m venv .venv	Create an isolated Python environment
az monitor metrics list ...	Retrieve a metric for operational evidence
git status --short	Confirm no secrets or generated private assessment files are staged

Assessment Flow and Support

106. Complete TRAQOM digital attendance.
107. Complete Assessment digital attendance.
108. Sit the 60-minute WA-SAQ, then the 60-minute PP.
109. Submit the two candidate papers on the LMS.
110. Sign the Assessment Summary Record.

Support: enquiry@tertiaryinfotech.com · +65 6100 0613 · <https://lms-tms.tertiaryinfotech.com/>